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Analysis of Martingale Strategy in Quantitative Trading Market Based on BiLSTM-Attention Model

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***Abstract*—In order to overcome the shortcomings of the traditional Martingale Strategy in the extreme market, such as the poor adaptability of the margin call and the fixed position adding mechanism, this study proposes an improved Martingale Strategy combining the BiLSTM-Attention model and the multi-dimensional dynamic management mechanism. By constructing a quantitative framework including data processing, time series prediction, strategy implementation and back testing evaluation, the BiLSTM-Attention model is used to capture the long-term time series dependence of Kweichow Moutai (600519) stock price, and generate price prediction signals for future trading days; It also designs a composite signal system, integrates RSI, MACD, volatility indicators and dynamic position management module, introduces multiple risk barriers, index gradient position adding mechanism and sliding window profit and loss stop strategy, and realizes risk adaptive control. The results of the back test show that in the extreme market from June to September 2024, the traditional Martingale Strategy has suffered a large loss, while the improved Martingale Strategy has avoided the loss caused by large-scale overweight by virtue of multi-dimensional technical indicators. The improved Martingale Strategy significantly reduces the loss risk in the trend market by filtering invalid signals through technical indicators, adjusting position size through volatility and active profit and loss stop mechanism, and verifies the effectiveness of the collaborative optimization of the deep learning model and traditional trading strategies. This framework provides an extensible technical path for low-frequency quantitative trading, and is suitable for personal lightweight financial product development.**

***Keywords*—BiLSTM-Attention, Martingale Strategy, quantitative trading**

I. INTRODUCTION

With its systematicness, discipline and efficiency, quantitative trading has become an indispensable investment tool in the financial market. As a classic position management method, Martingale Strategy shows its unique advantages in the volatile market through the reverse operation logic of "loss plus position and profit minus position". However, the "mean reversion" assumption it relies on is easy to cause the risk of capital margin call in extreme fluctuations or trend quotations, and the traditional fixed multiple position adding mechanism is also difficult to adapt to the complex market environment. How to optimize the risk control capability of Martingale Strategy has become an important research direction in the field of quantitative trading.

Early research on Martingale Strategy mainly focused on the field of traditional financial engineering. Owoloko et al. used Martingale Strategy to deeply analyze the probability theory phenomenon behind this strategy, proving that this seemingly simple "buy low and sell high" investment strategy actually has a solid mathematical foundation [1]. The traditional Martingale Strategy has limitations in fund management, and there is a risk of margin call in the unilateral market. To optimize this problem, Chen and Liao proposed an innovative Improved Martingale Betting Strategy (IMBS), introduced the stop loss mechanism and adjusted the betting doubling strategy [2].

At the same time, Machine Learning has made breakthroughs in time series prediction, especially the advantage of LSTM model in processing financial time series. Zou uses the LSTM model to capture the characteristics of the market at different time scales, so as to improve the accuracy of prediction [3]. Liu et al. proposed a lightweight LSTM method based on Singular Value Decomposition (SVD) to meet the challenge of deploying LSTM models in resource constrained environments, while maintaining high accuracy [4].

However, there are significant noises in financial market time series prediction, which will lead to deviation in the prediction results of traditional models. Although Deep Learning can process large-scale nonlinear data, it is often difficult for a single model to fully capture the internal patterns and features in the signal. Li et al. proposed an innovative Deep Learning hybrid prediction model CEEMDAN-Informer-LSTM for noise and uncertainty. This model effectively improves the prediction accuracy by combining CEEMDAN signal decomposition algorithm, Informer and LSTM models [5].

In recent years, LSTM has been widely used in stock trading systems. For instance, Zou et al. proposed an automated stock trading system based on Deep Reinforcement Learning (DRL) and Cascading Long Short-Term Memory (CLSTM). This system captures hidden information in daily stock data by leveraging a two-stage designed deep LSTM. The Proximal Policy Optimization (PPO) algorithm was adopted for training, effectively addressing the performance challenges of traditional DRL methods when dealing with low signal-to-noise ratio and unsmoothed financial data [6].

This study proposes an improved Martingale Strategy that integrates BiLSTM-Attention price prediction signals with multi-dimensional dynamic management. This study, by integrating Neural Network with trading strategies, has constructed a complete quantitative trading framework that encompasses data acquisition, model prediction, strategy execution, and effect evaluation.

II. Method

A. Data Source and Description

The daily stock line data is obtained through the akshare library interface. akshare is a third-party Python library dedicated to financial data acquisition, aiming to provide free and convenient financial data interfaces for quantitative trading, academic research, and more. It integrates multiple public data sources, covering areas such as stocks, funds, and futures, and particularly provides comprehensive data support for China's financial market. This study took Kweichow Moutai (600519) as the analysis target, with the time range set from January 1, 2018 to July 1, 2025. Complete daily line data of 1817 items in six fields were obtained, as shown in Table I.

TABLE I. Data Obtained from the Interface

Fields Name	Data Type	Description
date	(pandas)datetime	Dates
open	float64	Opening price of the day
high	float64	Highest price of the day
low	float64	Lowest price of the day
close	float64	Closing price of the day
volume	float64	Trading volume of the day

B. Data Preprocessing

Data preprocessing mainly focuses on the time series data of stock closing prices, aiming to provide standardized input for the training of the BiLSTM-Attention model. First, extract the closing price column from the stock dataset and convert it into a two-dimensional array to ensure that the data format meets the requirements for subsequent processing. Then, MinMaxScaler is used to normalize the data, scaling the original price to the range of [0,1] to eliminate the influence of dimensions and improve the efficiency of model training.

The normalized mathematical formula is:

$$X_{\text{scaled}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}} \tag{1}$$

Among them, $X_{\min}$ and $X_{\max}$ are the minimum and maximum values of the original data respectively.

Generate time series samples and corresponding labels through the sliding window. For normalized data of length N, samples are generated by sliding at a fixed time step (60 days) : each sample X_i contains continuous data from the i-th day to the $i+59$th day (i.e., $X_i = [X_i, X_{i+1}, \ldots, X_{i+59}]$), and the corresponding label Y_i is the closing price on the $i+60$th day ($Y_i = X_{i+60}$). This method converts unordered time series into structured supervised learning samples [7].

Finally, the generated samples are divided into a training set and a testing set in an 8:2 ratio, which are respectively used for model training and performance verification. If it is necessary to restore the normalized data to the original price, it can be achieved through the inverse transformation formula:

$$X_{\text{original}} = X_{\text{scaled}} \times (X_{\max} - X_{\min}) + X_{\min} \tag{2}$$

This preprocessing process ensures the standardization of the data and the adaptability of the model input, laying the foundation for the subsequent time series prediction tasks of the BiLSTM-Attention model.

C. BiLSTM-Attention Model

The BiLSTM-Attention model undertakes the core function of stock price prediction. By analyzing historical price sequences to predict future trends, the model receives the 60-day time window sequence data after preprocessing and outputs the predicted price value for the next trading day. This predicted value will be used by the Martingale Strategy to calculate the strength of long and short signals and serve as an important basis for trading decisions in backtesting.

The BiLSTM-Attention model is a Deep Learning model based on BiLSTM and the Attention mechanism, mainly used for handling prediction tasks of sequential data. As shown in Figure 1, the model first extracts the temporal features of the input sequence through the BiLSTM layer. BiLSTM can simultaneously capture the forward and backward information of a sequence, thereby providing a more comprehensive understanding of the contextual relationship of the data [8].

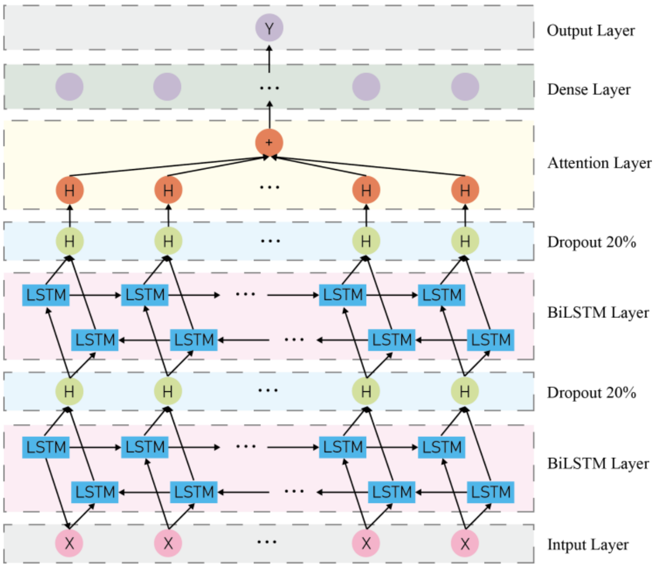

Fig. 1. BiLSTM-Attention network architecture

Forward LSTM calculation, processing sequence from t = 1 to t = T:

$$\overrightarrow{h_t} = \mathrm{LSTM}\left(x_t, \overrightarrow{h_{t-1}}\right) \tag{3}$$

Backward LSTM calculation, processing sequence from t = T to t = 1:

$$\overleftarrow{h_t} = \mathrm{LSTM}\left(x_t, \overleftarrow{h_{t+1}}\right) \tag{4}$$

Bidirectional splicing:

$$h_t = \left[\overrightarrow{h_t}; \overleftarrow{h_t}\right] \tag{5}$$

In the program, two BiLSTM layers are stacked successively, and a Dropout layer is added after each layer to prevent overfitting.

After the BiLSTM layer, the model introduces a custom Attention layer. This layer assigns different importance scores to the features at each time step of the LSTM output by learning trainable weights and biases [9]. The Attention mechanism first calculates the attention score at each time step:

$$e_t = \tanh(W \times h_t + b) \tag{6}$$

Among them, $W \times h_t$ projects the hidden state h_t onto the scalar space and calculates the original correlation score, and b add an independent bias term for each time step.

Then, through normalization, the attention weights for each time step are obtained:

$$\alpha_t = \frac{\exp(e_t)}{\sum_{i=1}^{T} \exp(e_i)} \tag{7}$$

Finally, multiply the input features by the weights and perform weighted summation to obtain a context vector:

$$c = \sum_{t=1}^{T} \alpha_t h_t \tag{8}$$

This ensures that the model can automatically focus on the time series segments that contribute most to the final prediction, thereby enhancing the model's expressive and generalization capabilities.

After the Attention layer, the model also includes a Dense layer and a Dropout layer for further feature extraction and regularization [10]. Finally, the final prediction result is generated through a linearly activated output layer: ds an independent bias term for each time step:

$$\begin{aligned} z &= \mathrm{ReLU}(W_z \times c + b_z) \\ \hat{y} &= W_o \times z + b_o \end{aligned} \tag{9}$$

The entire model is trained using the Adam optimizer and the Mean Square Error loss function, making it suitable for regression tasks.

D. Improved Matingale Strategy

The improved Matingale Strategy constructs a composite signal system, deeply integrating the RSI (Relative Strength Index) indicator with the MACD (Moving Average Convergence / Divergence) indicator. When calculating the signal strength, it is not only necessary to pay attention to whether the RSI has entered the overbought and oversold zone, but also to capture the momentum turning point through the changes in the MACD bar chart. When a golden cross (MACD line crossing the signal line) or a death cross (MACD line crossing the signal line crossing the signal line) occurs, a signal strength bonus in the corresponding direction will be given. At the same time, the BiLSTM-Attention model price prediction mechanism is introduced. When the predicted price deviates from the current price by more than 2%, additional signal weights are added. This multi-factor signal system effectively filters out false breakdowns and avoids blind position additions at the end of the trend.

The dynamic position management system has achieved adaptive risk adjustment. The volatility is calculated based on the standard deviation of the 20-day return rate, and the market environment is divided into three states: high volatility, medium volatility and low volatility. In a highly volatile environment, the position level is automatically reduced to 0.5 times the base position. For medium volatility, 0.75 times is adopted. Only when the volatility is low, the standard position is maintained. At the same time, the maximum capital usage limit for a single transaction is set at 50%, combined with the index position addition mechanism. This not only retains the loss compensation feature of the Martingale Strategy but also reduces the risk of margin call in extreme market.

Position calculation formula:

$$\mathrm{PositionSize}=\min\left(\mathrm{BasePosition} \times 2^{\mathrm{n}} \times \phi,\ 0.5 \times \mathrm{Capital}\right) \tag{10}$$

Among them, n represents the current number of positions held, and ϕ is the volatility adjustment coefficient:

$$\phi = \begin{cases} 0.5, \text{if } \delta > 0.03 \\ 1.0, \text{if } \delta < 0.02 \\ 0.75, \text{otherwise} \end{cases} \tag{11}$$

δ represents volatility, calculated through the standard deviation of price return:

$$\delta = \sqrt{\frac{1}{\mathrm{T}} \sum_{\mathrm{t}=1}^{\mathrm{T}} (\mathrm{r_t} - \bar{\mathrm{r}})^2}, \quad \mathrm{r_t} = \frac{\mathrm{p_t} - \mathrm{p_{t-1}}}{\mathrm{p_{t-1}}} \tag{12}$$

At the same time, a sliding window take-profit and stop-loss mechanism is introduced. Maintain the price history record for 5 trading days and calculate the cumulative return rate during the holding period in real time. When the yield reaches 4.5%, a take-profit is triggered; when the loss exceeds 3.5%, a forced stop-loss is imposed. Unlike the traditional Martingale Strategy's passive waiting for a pullback, this active exit mechanism can effectively lock in profits in trend markets and promptly cut losses in volatile markets.

At the level of opening positions logic, the strategy implements a triple risk barrier: 1) Volatility filtering, prohibiting the opening of new positions when the volatility exceeds 1.5 times the high threshold. 2) Direction consistency check, only additional positions in the same direction as existing positions are allowed. 3) Signal strength verification, long and short signals need to form an advantage ratio of more than 3:1 to trigger a trade.

Improve the Martingale Strategy by filtering out invalid signals through technical indicators, adjusting volatility to reduce extreme risks, and locking in profits through gradient profit-taking. The parameter system adopts a modular design and can adapt to different volatile market environments by adjusting the threshold. The code of this article comes from [11].

III. Results

A. BiLSTM-Attention Model Results

The training process of the BiLSTM-Attention model shows that the model converges rapidly and continuously optimizes, as shown in Figure 2. After 37 rounds of training (the early stop mechanism was triggered, the training was terminated if the loss did not improve after 5 consecutive rounds of verification), the final training loss was 0.002 and the verification loss was 0.0006. This phenomenon where the validation loss is significantly lower than the training loss indicates that the model demonstrates stronger generalization ability on unseen data, which may be attributed to the fact that the random perturbations introduced by the Dropout layer during training enhance the model's robustness.

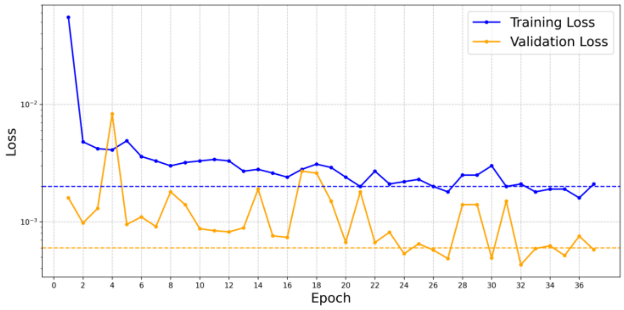

Fig. 2. The Loss curve of the BiLSTM-Attention model training process

The BiLSTM-Attention in the model architecture effectively captures the long-term dependency characteristics of stock price data. The effectiveness of this architecture for univariate time series prediction tasks was ultimately verified, laying a reliable foundation for subsequent stock price prediction and Martingale Strategy trading.

B. Trading Backtesting Results

Trading backtesting is carried out on the test set. From the perspective of the price prediction effect, as shown in Figure 3, the predicted price of the test set is basically consistent with the actual price trend, especially showing a high degree of following in trend market conditions. However, the prediction error for short-term fluctuations is slightly large and has obvious lag. This might be related to the limited ability of the BiLSTM-Attention model to capture short-term noise, but it verified the predictive effectiveness of the model as a whole.

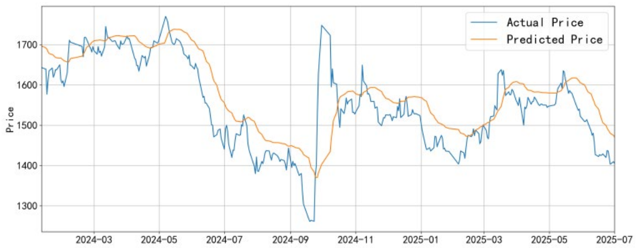

Fig. 3. Price forecast comparison

The improved Martingale Strategy comprehensively judges the opening conditions based on multiple indicators such as the predicted price signal of BiLSTM-Attention, RSI, MACD, and volatility, and implements dynamic take-profit and stop-loss through the sliding window yield rate. The backtest log shows that the system executed a total of 84 trading operations, including 28 opening positions (22 opening long positions and 6 opening short positions), 23 adding positions (19 adding long positions and 4 adding short positions), 6 reducing positions (all stop-loss reduction positions), and 27 closing positions (22 stop-loss closing positions and 5 stop-profit closing positions). Trading behavior is mainly concentrated during periods with low volatility (<2.5%) and clear technical indicators (RSI<30 or >70, MACD golden cross or death cross), which is in line with the logic of strategy design.

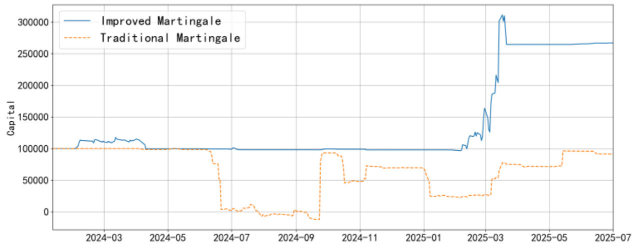

Fig. 4. Changes in account funds

The improved Martingale Strategy's capital curve shows a phased growth feature, while the traditional Martingale Strategy's capital only fluctuates below the cost line, as shown in Figure 4. The improved Martingale Strategy initially experiences a slight pullback due to adapting to market fluctuations, but as the strategy captures the trend, the capital scale rises phased. The final total return rate was 166.99%, and the Sharpe ratio was 1.39, indicating an excellent return level per unit risk. The maximum drawdown of 15.01% mainly occurs in extreme market conditions where market volatility suddenly rises (>3.5%), triggering the stop-loss mechanism of the strategy and significantly improving the deficiency of expanded losses in the traditional Martingale Strategy.

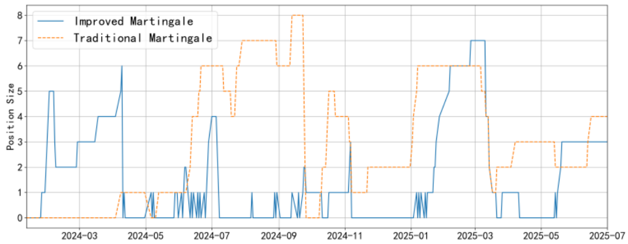

Fig. 5. Changes in the number of positions held

The changes in the number of positions held by different strategies are shown in Figure 5. The improved Martingale Strategy mostly has 1 to 3 positions held, and only triggers additional positions when there are continuous reverse fluctuations (up to 7 levels), which conforms to the core logic of the Martingale Strategy of "adding positions against the trend". By comparing different strategies in Figure 5, during the rapid decline of stock prices from June to September 2024, the traditional Martingale Strategy made quick purchases, while the improved Martingale Strategy, in combination with technical indicators, did not make large-scale increases in holdings, thus avoiding losses in a downtrend market. The improved Martingale Strategy, combined with the technical indicator visualization charts in Figures 6, 7, 8, and 9, is highly consistent with the RSI oversold/overbought signals and the opening operation. The positive and negative conversion of the MACD bar chart effectively assists in trend judgment, and the dynamic adjustment of the position size by the volatility threshold (2.5%-3.5%) also significantly reduces the trading risk in a high-volatility environment.

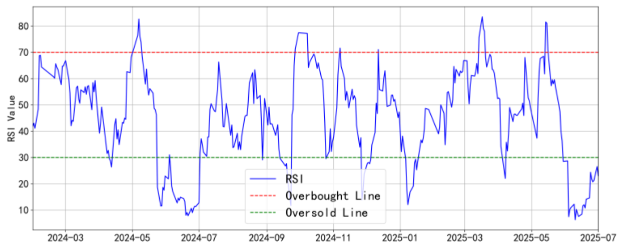

Fig. 6. RSI indicator

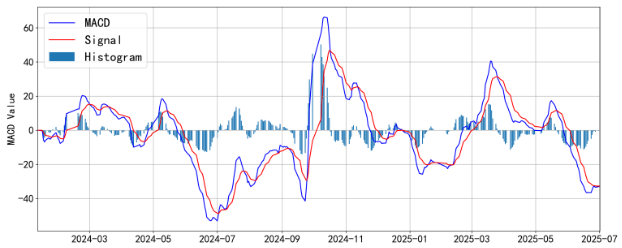

Fig. 7. MACD indicator

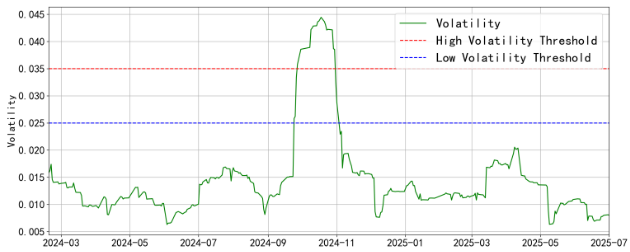

Fig. 8. Price volatility indicator

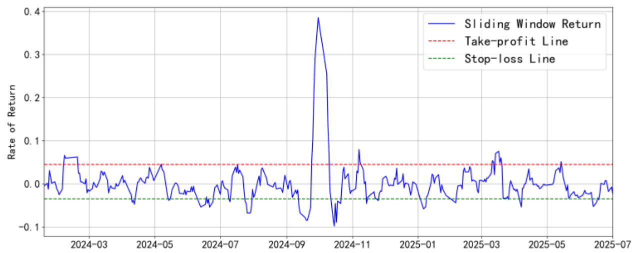

Fig. 9. Sliding window return rate indicator

The improved Martingale Strategy demonstrated stable profitability and risk control capabilities during the backtesting period. The BiLSTM-Attention prediction model provided effective trend guidance for the strategy, and the dynamic position management of the Martingale Strategy further optimized the return-risk ratio.

IV. Conclusion

This study successfully constructed an improved Martingale Strategy quantitative trading framework by integrating BiLSTM-Attention price prediction and multi-dimensional dynamic management mechanisms. Experiments show that this framework effectively improves the inherent flaws of the traditional Martingale Strategy in extreme market conditions: The BiLSTM-Attention model provides predictive signals for stock prices and reliable direction guidance for the strategy; A multi-factor signal system that simultaneously integrates RSI, MACD, volatility and sliding window return rate indicators, combined with a dynamic position adjustment mechanism, has enhanced the level of risk control.

This study confirms that the collaborative innovation of Deep Learning methods and traditional trading strategies has significant potential: the time series modeling ability of BiLSTM-Attention makes up for the shortcoming of trend judgment in the Martingale Strategy, while the dynamic risk management module realizes the adaptation of the strategy to the market environment through a parametric system. This study focuses on a single model and specific individual stocks. Future research can further explore the application effects of Ensemble Learning, Model Fusion, or hybrid architectures (such as CNN-LSTM, Transformer-Attention) in more complex financial scenarios such as capturing a broader market and processing higher-dimensional data to enhance the robustness of predictions and the adaptability of strategies. This research provides scalable technical references for the field of low-frequency quantitative trading and is suitable for the strategy development of lightweight personal financial products.

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